



OPEN A dynamic QCA of new quality productivity driving high quality economic development in the yellow river basin

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The high-quality economic development (HQED) of the Yellow River Basin (YRB) faces dual constraints of growth momentum transformation and regional imbalance. New Quality Productive Forces (NQP) are regarded as a key breakthrough, yet the complex causal mechanisms through which they drive HQED remain unclear. Traditional research methods fall short in uncovering the multifactorial, concurrent causal relationships and the spatio-temporal dynamics involved. Therefore, drawing on panel data from 49 prefecture-level cities in the YRB from 2012 to 2022, this study employs the CRITIC–TOPSIS method and dynamic QCA to measure the HQED index and to explore in depth the configurational pathways and dynamic processes through which NQP evolution fosters the emergence of HQED. The findings are as follows: (1) During the study period, the HQED index of the YRB exhibited a “W-shaped” evolutionary trend, with a spatial gradient of “higher in the east and lower in the west.” (2) No single NQP factor emerged as a necessary condition for HQED. (3) The pathways to HQED in the YRB can be categorized into three models: the single-core innovation-driven model, the region-wide innovation resource synergy-driven model, and the comprehensive digital transformation-driven model. Technological innovation plays a universal role, but its effectiveness requires synergy with other factors to be fully realized. (4) While no obvious temporal effects were observed in the configurational pathways, significant spatial heterogeneity exists, with different urban agglomerations displaying distinct pathway preferences. This study reveals the complex interactive mechanisms through which NQP drives HQED in the YRB and provides refined policy implications for place-based and coordinated regional development.

Keywords New quality productive forces, High-quality economic development, Yellow river basin, Dynamic QCA

Since the Industrial Revolution, the world has undergone an unprecedented wave of urbanization¹. Cities, as the hubs of economic activity and population concentration, now house over 50% of the global population and account for 75% of global energy consumption². However, the strain on resources and the environment caused by the agglomeration of people and industries³, regional development disparities and governance failures^{4,5}, and the innovation bottlenecks inherent in traditional growth models⁶, together present the fundamental challenges restricting urban development. Therefore, merely focusing on expanding total economic output is insufficient to meet the demands of sustainable development. Enhancing the quality and efficiency of urban economic development is crucial for addressing these challenges.

Since becoming the largest developing country in the world, China’s urbanization rate has risen to 67%, an increase of 55.52 percentage points compared to the end of 1949⁷. Cities have become the main engine of China’s economic growth and an important vehicle for modernization. The report of the 19th National Congress of the Communist Party of China pointed out that China’s economy has shifted from a stage of rapid growth to a stage of high-quality development⁸. This means that the development of urban economies in China must also transition from quantitative expansion to qualitative improvement⁹. The Yellow River Basin (YRB) is an important ecological barrier and economic zone in China, yet its urban development faces unique challenges, such as the complex water–sand relationship¹⁰, slow industrial chain upgrading¹¹, and regional development imbalance¹². As a result, in 2021, the ecological protection and high-quality development of the YRB were

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designated as a national strategy¹³. A key issue that urgently needs to be addressed is how to promote high-quality economic development (HQED) in cities within the basin while preserving the ecological environment of the YRB.

For the YRB, the traditional urban development model based on resource consumption and scale expansion is no longer sustainable, and there is an urgent need to reconstruct the development path with innovative thinking¹⁴. New Quality Productive Forces (NQP) refer to a new form of productivity driven by digital technologies, knowledge capital, and collaborative innovation, integrating features of intelligence, data, and ecology¹⁵. Developing NQP is a response to the current overall pattern of China's economic development and serves as an important engine for promoting HQED in the YRB^{16,17}. Against this backdrop, this paper explores the impact of NQP on HQED in the YRB and the complex causal mechanisms involved, offering policy recommendations to achieve HQED in the YRB.

Internationally, academic research on NQP and HQED has mainly focused on three areas:

First, the conceptual interpretation of NQP. The notion of NQP was formally introduced by President Xi Jinping in 2023 and has been defined in authoritative sources as advanced productive forces driven primarily by innovation¹⁸. Within the academic community, NQP is widely regarded as both an extension and a contemporary development of the Marxist theory of productive forces in the Chinese context. It inherits the core principles of classical Marxist productive force theory, incorporates insights from the theory of technological innovation, and is closely aligned with the framework of a modern industrial system. Conceptually, NQP encompasses three interrelated components: a highly skilled workforce, new forms of means of labor, and new types of objects of labor. It represents an advanced stage of productive forces, whose essential connotation lies in the qualitative transformation resulting from the optimized integration of these elements^{19–21}. Understanding NQP requires emphasizing the theoretical extension of its connotation: the term “new” refers to the emergence of novel production factors and innovative modes of their combination; “quality” reflects a robust industrial foundation and sustainable development momentum; and “force” extends beyond traditional material productive forces to incorporate emerging elements such as digital technologies, knowledge capital, and green, low-carbon capabilities^{22,23}. Nonetheless, most existing studies offer qualitative characterizations of NQP, and the development of a systematic analytical framework for its study remains an important area for ongoing research.

Second, the interpretation of the internal logic through which NQP promotes HQED. Scholars generally view HQED as a gradual process, and related discussions are closely aligned with the broader academic discourse on sustainable development and inclusive growth. Factors such as the allocation of innovation elements²⁴, industrial structure upgrading²⁵, environmental regulation²⁶, institutional environment²⁷, and digital finance²⁸ are widely recognized as important drivers of HQED. Existing studies suggest that NQP can overcome problems inherent in traditional productivity models, such as resource depletion and environmental pollution, and instead drive regional high-quality economic development through multiple channels, including talent agglomeration²⁹, deepening of scientific and technological innovation mechanisms²⁶, construction of new forms of digital infrastructure³⁰, optimization of factor resource allocation³¹, and integration of green development³².

Third, with respect to the mechanisms through which NQP drives HQED, systematic research in the current literature remains in the early stage of theoretical inquiry, and empirical evidence is relatively limited. Notably, related research domains have produced methodologically instructive findings. Building on dimensions such as high-tech industry agglomeration, allocation of innovation resources, digital inclusive finance, and green finance innovation, scholars have employed empirical approaches—including the spatial Durbin model³³, threshold effect model²⁷, mediation model³⁴, and difference-in-differences model³⁵—to examine underlying mechanisms, driving effects, threshold effects, spatial spillovers, and patterns of regional heterogeneity. Nevertheless, this “net effect”-oriented paradigm has a fundamental methodological limitation: it tends to decompose complex drivers into independent variables and assess their marginal contributions in isolation, which is fundamentally at odds with the systemic and integrative nature of NQP. To address this limitation, some scholars have introduced Qualitative Comparative Analysis (QCA), which shifts the analytical focus from the independent impact of single variables to the identification of multiple configurational pathways capable of producing the same outcome³⁶. In the social sciences, QCA has been applied to investigate topics such as digital transformation performance³⁷, carbon emissions performance³⁸, green infrastructure development³⁹, and complex urban transitions⁴⁰. The multi-factor interplay through which NQP influences HQED unfolds as a temporally continuous process, rendering static, cross-sectional configurations in traditional QCA inadequate for capturing the intricate conjunctural causality and the spatio-temporal interactions among factors.

In summary, substantial gaps remain in existing research. First, the academic literature on the NQP–HQED nexus is sparse, with most contributions situated at the theoretical level and offering little empirical validation. Although related fields have developed analytical frameworks for HQED, these frameworks largely remain anchored in the traditional dimensions of productive force evolution. Second, studies on HQED drivers have generally examined single factors using conventional regression methods, thereby neglecting the complex conjunctural causality arising from multiple concurrent factors. Third, existing QCA-based studies have primarily addressed the static relationships between antecedents and outcomes, overlooking temporal effects and regional heterogeneity, which limits the capacity of their findings to capture the dynamic nature of the phenomenon.

The contributions of this study are reflected in three main aspects. First, it moves beyond the traditional “three-element” framework of productive force theory by integrating the multidimensional characteristics of

NQP to construct a “technology–green–digital” theoretical model, thereby systematically explaining the complex causal mechanisms through which NQP influences HQED. Second, by measuring the HQED index of the YRB from 2012 to 2022 using the CRITIC–TOPSIS method and introducing a dynamic QCA approach to panel data analysis, this study not only identifies the static configurational pathways of multi-factor linkages in NQP but also uncovers their temporal dynamics and spatial heterogeneity, thus overcoming the “spatio-temporal blind spots” inherent in conventional QCA. Finally, it provides a theoretical foundation for coordinating “technology–green–digital” development strategies in the YRB.

This paper is structured into six sections. The first section presents the research background and reviews the current state of relevant studies. The second section develops the theoretical model of NQP and elaborates on the background and theoretical framework for its influence on HQED in the YRB. The third section outlines the methodological approach. The fourth section reports the empirical analysis and discussion. The fifth section offers conclusions and policy recommendations, while also identifying research limitations and directions for future work.

Background and theoretical framework

The dual goals of ecological protection and high-quality development in the YRB place new demands on regional development theory. This study focuses on NQP as an emerging paradigm, conceptualizing it as a complex system that integrates technological innovation, green development, and digital transformation. On this basis, a theoretical analytical framework is constructed to systematically reveal how NQP empowers HQED in the YRB.

Theoretical framework

To gain deeper insight into the complex mechanisms through which NQP drives HQED, this study draws on classical academic perspectives such as complex systems theory⁴¹ and endogenous growth theory⁴², using them as the theoretical foundation for constructing the analytical framework. As the core driving force of HQED, the development of NQP itself also requires breakthroughs beyond traditional frameworks. Current scholarly understanding of NQP largely focuses on its multidimensional and multi-level characteristics, yet certain points of consensus have emerged. First, NQP, like traditional productive forces, constitutes a vital engine for economic growth and industrial upgrading; however, its underlying drivers and realization pathways differ fundamentally from those of traditional productive forces and should be reinterpreted from the perspective of modern productive force theory¹⁸. Compared with its traditional counterpart, NQP encompasses not only advancements in the sphere of material production but also qualitative transformations across social, environmental, and technological dimensions. Accordingly, the NQP model should break through the classical Marxist tripartite framework of productive forces—namely, laborers, means of labor, and objects of labor⁴³. Second, NQP originates from the innovation and further development of the traditional productive force concept, referring to advanced productive forces formed through the interaction of production factors in domains such as technological innovation, green development, and digitalization⁴⁴. Building on this understanding, the present study develops a “technology–green–digital” three-dimensional theoretical model of NQP and, from a complex systems perspective, constructs a theoretical framework elucidating the mechanisms through which NQP influences HQED, as illustrated in Fig. 1.

Technological productivity (TP) and its impact on HQED

At the core of NQP lies the driving role of technological innovation, with a focus on enhancing production efficiency and quality through scientific and technological advances. According to Endogenous Growth Theory, technological progress and knowledge accumulation are endogenous drivers of long-term economic growth rather than mere external factor inputs⁴⁵. Technological productivity (TP) facilitates economic transformation and upgrading, optimizes industrial structures, and improves production efficiency. By stimulating innovative vitality, deepening R&D activities, and accelerating the commercialization of research outcomes, TP provides efficient technological support for every link of the industrial chain while optimizing the allocation of production factors, thereby promoting automation and intelligent production processes. TP also enhances the innovation capacity and technological sophistication of enterprises, strengthening the resilience of the economic system and

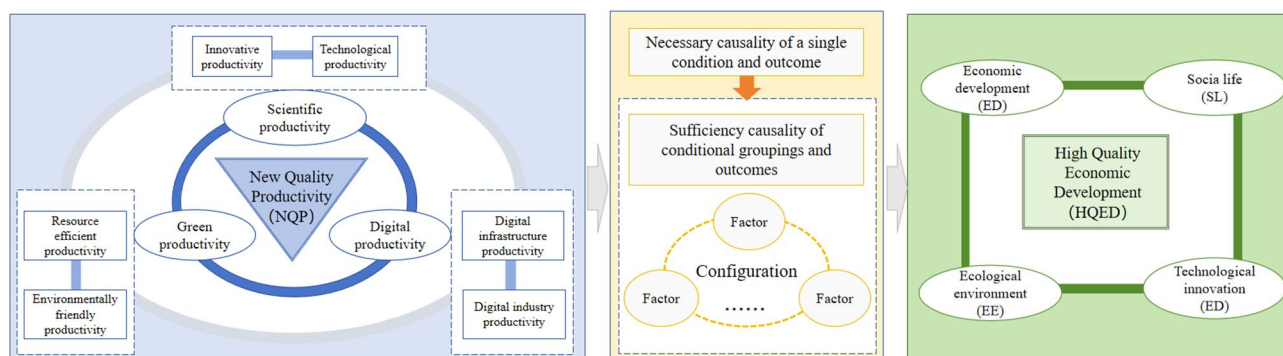


Fig. 1. Logical Framework for NQP’s Multi-Factor Linkage to Impact HQED.

enabling it to play a pivotal role in achieving high-quality development. In this study, innovation productivity (X1) and technology productivity (X2) are selected to measure TP.

Green productivity (GP) and its impact on HQED

NQP emphasizes the coordinated development of the economy and the environment. Both Ecological Modernization Theory and the Porter Hypothesis posit that well-designed environmental regulations can stimulate innovation and achieve a “win–win” outcome for the economy and the environment^{46,47}. Green productivity (GP) enhances resource-use efficiency and reduces pollutant emissions, thereby generating both economic and ecological benefits. It guides the transformation of traditional industries toward low-carbon and environmentally friendly trajectories, improves the comprehensive utilization of resources, and reduces dependence on natural resources, thus providing new momentum for HQED. In this study, environmentally friendly productivity (X3) and resource-saving productivity (X4) are adopted as measures of GP.

Digital productivity (DP) and its impact on HQED

NQP underscores the broad application of digital technologies. Goldfarb et al. (2019) identify digital technology as a general-purpose technology characterized by zero marginal costs and pronounced network effects⁴⁸. Leveraging these characteristics, digital productivity (DP) improves the efficiency of information flows through robust digital infrastructure and, via the multiplier effect of data as a production factor, enables disruptive restructuring of resource allocation and industrial models across society. This process spawns new industries and business models, injecting intelligent and precise new drivers into HQED. Accordingly, this study employs digital infrastructure productivity (X5) and digital industry productivity (X6) to measure this dimension.

On this basis, the sampling dataset for this study was established. The YRB was selected because its dual policy priorities of “ecological protection” and “high-quality development” align closely with the “technology–green–digital” synergy framework proposed in this research, while the internal heterogeneity among its cities offers the case diversity necessary for QCA to identify multiple configurational pathways. The period from 2012 to 2022 was chosen as the study horizon, providing the empirical foundation for the dynamic analyses conducted in this work.

Indicator system

The selection of indicators should reflect the intrinsic logic and core characteristics of NQP and HQED. This paper measures the outcome variables from four dimensions: Economic Development (ED), Social Life (SL), Ecological Environment (EE), and Technological Innovation (TI). Based on the concept and connotation of NQP and the “Technology–Green–Digital” three-dimensional theoretical model, six dimensions are selected as the conditioning variables: Innovation Productivity (X1), Technological Productivity (X2), Environmentally Friendly Productivity (X3), Resource-Saving Productivity (X4), Digital Infrastructure Productivity (X5), and Digital Industry Productivity (X6), as shown in Table. 1.

Material and methods

The YRB constitutes a vital ecological barrier and economic zone in China, encompassing the Central Plains Urban Agglomeration, Jinzhong Urban Agglomeration, Lanxi Urban Agglomeration, Guanzhong Plain Urban Agglomeration, Ningxia Along-the-Yellow Urban Agglomeration, Shandong Peninsula Urban Agglomeration, and Hohhot–Baotou–Ordos–Yulin (Hubaoeyu) Urban Agglomeration. Spanning the upper, middle, and lower reaches of the basin, these seven major urban agglomerations exhibit significant heterogeneity in resource endowments, industrial structures, and stages of development, thereby providing a rich “natural experiment” sample for employing dynamic QCA to explore diverse pathways toward high-quality development. Based on this, and following the classification criteria outlined in the 2016 China Urban Agglomeration Development Report, this study selects 49 prefecture-level cities within the seven YRB urban agglomerations as the research sample (see Fig. 2). The study period is set from 2012 to 2022. The required data are primarily drawn from the *China Urban Statistical Yearbook* (2013–2023), *China Statistical Yearbook*, *China Science and Technology Statistical Yearbook*, *China Labor Statistical Yearbook*, the statistical yearbooks of the relevant provinces (municipalities/autonomous regions), the *China E-Commerce Report*, and the China National Intellectual Property Administration. Missing data are supplemented using linear interpolation and analogy methods.

For the quantification of HQED, this study adopts the CRITIC–TOPSIS method for comprehensive evaluation. HQED is a multidimensional composite indicator system, and its measurement requires addressing the objectivity of weight determination for each indicator. In this study, the CRITIC method is used to determine the weights. Its advantage lies in the fact that it not only measures the internal informational variability of an indicator through its standard deviation but also evaluates its conflict with other indicators by calculating correlations. An indicator with low correlation (i.e., low conflict) with others but high variability is assigned a higher weight, thereby effectively avoiding distortion caused by overlapping information among indicators. After obtaining objective weights, the TOPSIS method is applied for comprehensive measurement. The core idea of TOPSIS is to rank evaluation objects by calculating their relative distances to the “optimal ideal solution” and the “worst ideal solution.” The closer a city is to the optimal solution and the farther it is from the worst solution, the higher its HQED level. The integration of CRITIC and TOPSIS thus constructs a fully objective measurement system, spanning from weight determination to comprehensive evaluation, ensuring the scientific validity and credibility of the resulting HQED index⁴⁹. The specific steps are as follows:

(1) Indicator dimensionless processing

Positive indicators:

Goal Layer	Criterion Layer	Index	Calculations	Index Attribute
High-quality economic development (HQED)	Economic development(ED)	Inclusive TFP (A1)	Inclusive TFP index	+
		Advanced industrial structure (A2)	Tertiary sector output/secondary sector output	+
		Rationalization of industrial structure (A3)	Thiel coefficient for number of persons employed in the three industries/value of production	-
		Share of productive services (A4)	Number of employees in productive services/employees in urban units	+
	Social life(SL)	Personnel upgrading inputs (B1)	Expenditure on education/urban resident population	+
		Medical resource inputs (B2)	Number of hospital beds/urban resident population	+
		Unemployment rate of registered urban population (B3)	Unemployed persons/ [unemployed persons + (establishments + private, individual workers)]	-
		Per capita GDP (B4)	Gross Domestic Product (GDP) \ city Population	+
	Ecological environment(EE)	Sulfur dioxide removal (C1)	(Sulphur dioxide production—sulphur dioxide emissions)/sulphur dioxide production	+
		Rate of non-hazardous treatment of domestic waste (C2)	Non-hazardous treatment rate of domestic waste	+
		Annual average concentration of respirable fine particulate matter (C3)	Annual average concentration of respirable fine particulate matter	-
	Technological innovation(TI)	Level of innovation (D1)	Innovation index	+
New quality productivity(NQP)	Technological productivity(TP)	RD funding intensity (D2) Innovation productivity (X1)	R&D expenditure/GDP	+
			Patent grants per 10,000 people	+
			Number of RD workers/total employed population	+
		Technology productivity (X2)	Science and technology expenditure/local general public budget expenditure	+
			Number of persons employed in scientific research and technical services and geological prospecting/total employed population	+
			Robot mounted raw density	+
	Green Productivity(GP)	Resource-efficient productivity (X3)	Total energy consumption/gross regional product	-
			Per capita domestic water consumption	-
		Environmentally friendly productivity (X4)	Comprehensive utilization rate of general industrial solid waste	+
			Industrial wastewater discharge/gross regional product	-
	Digital productivity(DP)	Digital infrastructure productivity (X5)	Industrial sulphur dioxide emissions/gross regional product	-
			Number of Internet broadband access subscribers/total regional population*100	+
			Length of long-distance fiber-optic cable lines/area of administrative region	+
		Digital industry productivity (X6)	Number of cell phone subscribers/total regional population*100	+
			Number of persons employed in information transmission computer services and software/total employed population	+
			Revenue from telecommunication services/gross regional product	+
			E-commerce turnover/gross regional product	+

Table 1. Illustrative table of NQP and HQED indicators.

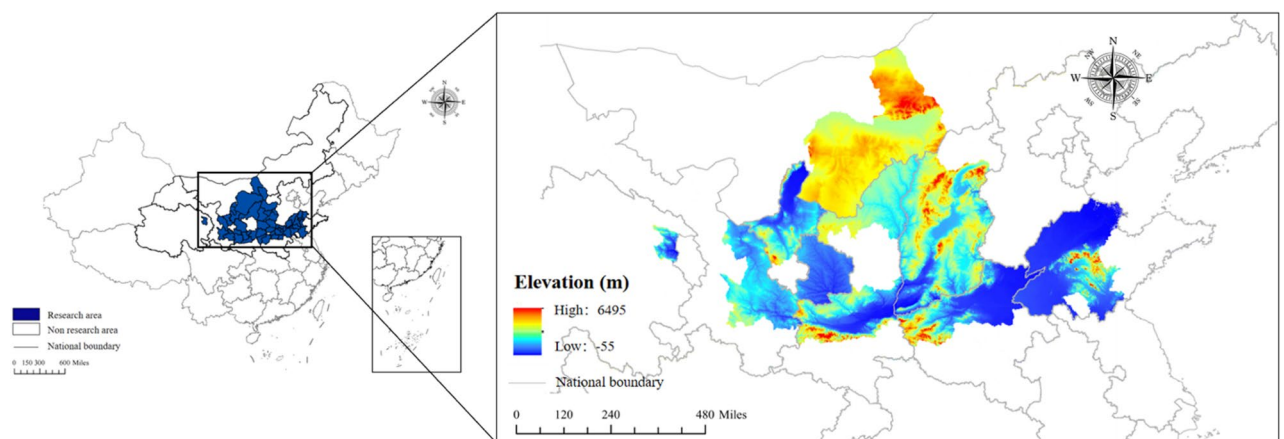


Fig. 2. Overview of the study area.

$$X_{ij} = \frac{x_{ij} - \min(x_{1j}, x_{2j}, \dots, x_{nj})}{\max(x_{1j}, x_{2j}, \dots, x_{nj}) - \min(x_{1j}, x_{2j}, \dots, x_{nj})} \quad (1)$$

Negative indicators:

$$X_{ij} = \frac{\max(x_{1j}, x_{2j}, \dots, x_{nj}) - x_{ij}}{\max(x_{1j}, x_{2j}, \dots, x_{nj}) - \min(x_{1j}, x_{2j}, \dots, x_{nj})} \quad (2)$$

In this context, x_{ij} represents the raw value of the j -th indicator for the i -th city, while X_{ij} denotes the standardized value of the j -th indicator for the i -th city.

(2) Indicator weights

$$W_j = \frac{C_j}{\sum_{j=1}^n C_j} = \frac{S_j \times R_j}{\sum_{j=1}^n (S_j \times R_j)} = \frac{\sqrt{\frac{\sum_{i=1}^m (x_{ij} - \bar{x}_j)^2}{m-1}} \times \sum_{j=1}^n 1 - r_{jk}}{\sum_{j=1}^n \left(\sqrt{\frac{\sum_{i=1}^m (x_{ij} - \bar{x}_j)^2}{m-1}} \times \sum_{j=1}^n 1 - r_{jk} \right)} \quad (3)$$

where r_{jk} is the correlation coefficient between the j -th indicator and the k -th indicator; S_j is the variability indicator, R_j is the conflict indicator, C_j is the information carrying capacity; and W_j is the indicator weight.

(3) Positive and negative ideal solutions

$$Z^+ = \{ \max(X_{11}, X_{21}, \dots, X_{i1}), \dots, \max(X_{1j}, X_{2j}, \dots, X_{ij}) \} \quad (4)$$

$$Z^- = \{ \min(X_{11}, X_{21}, \dots, X_{i1}), \dots, \min(X_{1j}, X_{2j}, \dots, X_{ij}) \} \quad (5)$$

(4) Calculation of the HQED index

$$S_i = \frac{\sqrt{\sum_{i=1}^n W_j (X_{ij} - Z^-)^2}}{\sqrt{\sum_{i=1}^n W_j (X_{ij} - Z^+)^2} + \sqrt{\sum_{i=1}^n W_j (X_{ij} - Z^-)^2}} \quad (6)$$

where S_i is the HQED level of YRB.

This study adopts the Qualitative Comparative Analysis (QCA) method, a set-theoretic, case-oriented configurational approach. QCA acknowledges the complexity of socio-economic phenomena, with its core strength lying in its ability to identify different combinations of conditions that lead to the same outcome. However, the driving effect of NQP on HQED is not static; rather, it is a dynamic process that evolves continuously over time and exhibits spatial heterogeneity. Traditional QCA, which is mostly applied to cross-sectional data, has limited capacity to explain how complex causal relationships among multiple factors interact and evolve across temporal and spatial dimensions. To overcome this limitation, this study draws on Garcia-Castro et al. (2016) and employs dynamic QCA to analyze panel data⁵⁰. By systematically assessing the consistency and coverage of specific configurations across different time periods and geographic clusters, this method can precisely identify whether the multiple configurational pathways driving HQED are universal, time-dependent, or characterized by significant regional specificity. This enables us to move beyond the “spatio-temporal blind spots” of traditional QCA and conduct a systematic examination of the dynamic evolution and heterogeneous features of these configurational pathways across temporal and spatial dimensions.

Analysis, results and discussion

YRB's HQED evaluation of spatio-temporal evolution

Time-series characteristics of YRB's HQED

Using the CRITIC-TOPSIS method, this study calculates the HQED index of the YRB for the period 2012–2022, as shown in Fig. 3a. Overall, the index values are mainly distributed between 0.35 and 0.65, with the median remaining stable at approximately 0.50, indicating considerable room for improvement. Unlike the consistently upward trajectories observed in studies of coastal, economically developed urban agglomerations such as the Beijing–Tianjin–Hebei, Yangtze River Delta, and Pearl River Delta^{51,52}, the temporal evolution of the YRB's HQED index exhibits a “W”-shaped pattern, which can be divided into three distinct stages: an initial phase (2012–2015) of steady improvement with relatively small regional disparities; a middle phase (2016–2018) reaching a peak but accompanied by intensified internal imbalances; and a later phase (2019–2022) of slight decline followed by stabilization, during which regional disparities narrowed. A small number of outliers are present in certain years (e.g., 2016 and 2018). Compared with coastal regions, these results underscore the structural vulnerabilities characteristic of the YRB as a typical inland river basin during China's economic transition period.

Based on the four-dimensional HQED index data for the YRB from 2012 to 2022, as shown in Fig. 3b, the region's development demonstrates pronounced structural imbalance and dynamic evolution. The SL dimension

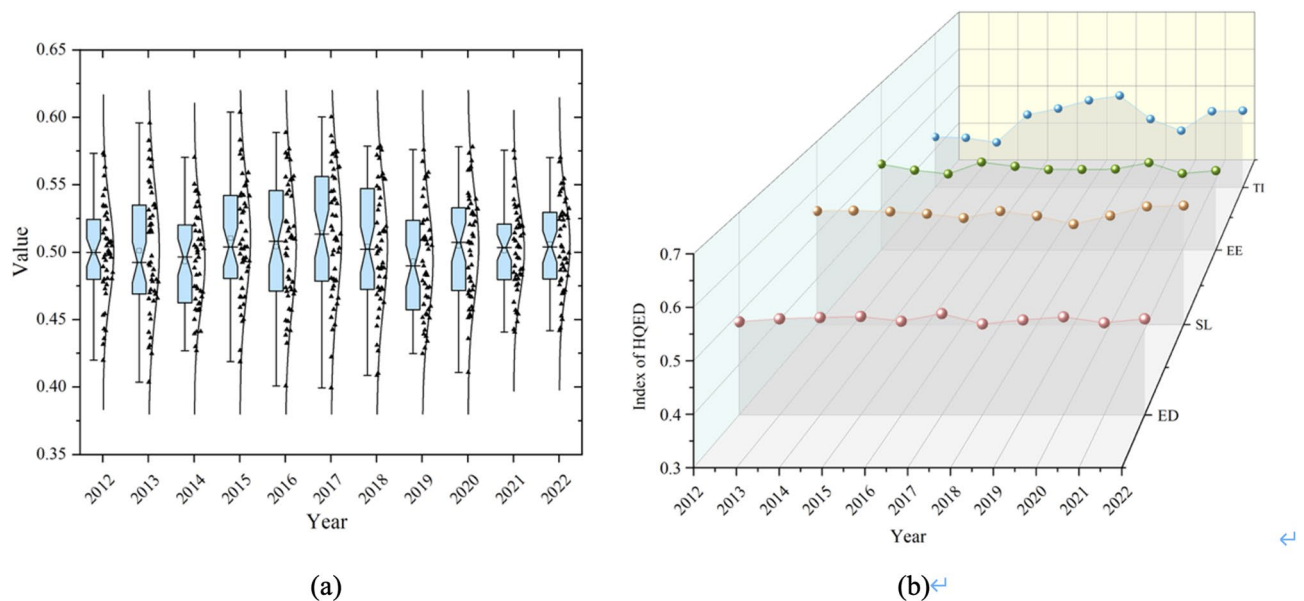


Fig. 3. Distribution of the HQED index. Note: (a) (b) denote the time series plots of HQED and its sub-dimension indices of YRB respectively.

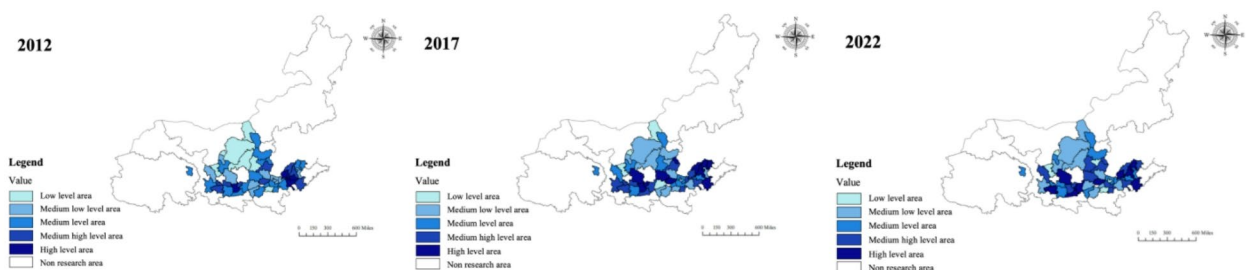


Fig. 4. Temporal evolution of the HQED index.

performed best and has strengthened steadily in recent years, reflecting significant achievements in improving livelihoods. The EE index fluctuated, suggesting that the effectiveness of environmental governance remains unstable. The ED index has stagnated over the long term, becoming a bottleneck for overall development. Notably, although the TI index experienced substantial growth, it did not effectively drive a corresponding improvement in economic development quality. While previous studies have confirmed the role of innovation in promoting HQED⁵³, the present findings suggest that in the YRB there may be obstacles in the mechanism that translates innovation outcomes into real productive forces. Consistent with the research of Lu et al.¹⁴, the YRB should focus on strengthening the effective integration of technological innovation and economic development, consolidating its advantages in social development, enhancing long-term mechanisms for environmental governance, and promoting multidimensional synergy through systematic planning to achieve structural optimization and momentum transformation in HQED.

Spatial and temporal evolution of HQED in YRB

The spatial distribution of the YRB's HQED index in 2012, 2017, and 2022, as shown in Fig. 4, reveals pronounced spatio-temporal evolution characteristics. From a spatial perspective, the HQED in the YRB exhibits a distinct “high in the lower reaches, low in the upper and middle reaches” gradient pattern. High-value and medium-high-value areas are primarily concentrated in the middle and lower reaches of the basin, particularly in the core areas of provinces such as Henan and Shandong, forming a “point-block” spatial agglomeration pattern. The upper reaches are predominantly characterized by low-value areas, indicating a spatial polarization of development within the YRB. From the temporal evolution perspective, between 2012 and 2022, high-value and medium-high-value areas generally showed a contraction trend. Cities with relatively high HQED levels often displayed a “patch-like” distribution rather than evolving into contiguous high-development zones, suggesting that the “trickle-down effect” within the region is not significant and that the capacity of core cities to generate radiating, driving effects requires strengthening. This finding provides an important reflection on the applicability of the traditional growth pole theory to the YRB⁵⁴ and underscores the importance of breaking administrative barriers

		Full Membership	Crossover Point	Full Non-membership
Outcome Variables	New quality productivity(Y)	0.534	0.503	0.472
Conditional Variables	Innovation productivity (X1)	0.622	0.460	0.344
	Technology productivity (X2)	0.272	0.192	0.124
	Resource-efficient productivity (X3)	0.900	0.841	0.759
	Environmentally friendly productivity (X4)	0.678	0.634	0.570
	Digital infrastructure productivity (X5)	0.331	0.215	0.151
	Digital industry productivity (X6)	0.203	0.121	0.080

Table. 2. Calibration results for conditional and outcome variables.

Conditional Variables	Y				~ Y			
	Aggregate consistency	Aggregate coverage	Inter-group consistency adjustment distance	Intra-group consistency adjustment distance	Aggregate consistency	Aggregate coverage	Inter-group consistency adjustment distance	Intra-group consistency adjustment distance
X ₁	0.709	0.714	0.113	0.566	0.379	0.369	0.240	0.729
~ X ₁	0.373	0.383	0.211	0.778	0.706	0.701	0.131	0.658
X ₂	0.526	0.526	0.414	0.559	0.573	0.555	0.323	0.580
~ X ₂	0.555	0.573	0.385	0.615	0.511	0.511	0.360	0.587
X ₃	0.537	0.56	0.218	0.332	0.524	0.529	0.218	0.361
~ X ₃	0.548	0.543	0.218	0.311	0.564	0.541	0.193	0.297
X ₄	0.548	0.564	0.178	0.318	0.523	0.521	0.196	0.334
~ X ₄	0.535	0.537	0.182	0.332	0.562	0.546	0.178	0.290
X ₅	0.506	0.503	0.3632	0.630	0.609	0.585	0.331	0.580
~ X ₅	0.582	0.606	0.345	0.559	0.483	0.486	0.454	0.594
X ₆	0.568	0.568	0.360	0.538	0.548	0.531	0.334	0.545
~ X ₆	0.531	0.548	0.389	0.509	0.554	0.554	0.334	0.516

Table. 3. Results of the necessity analysis.

and promoting coordinated development among urban agglomerations to achieve high-quality development across the entire YRB.

YRB’s HQED configuration path analysis

Data calibration

This paper uses the direct calibration method to standardize the data, enabling subsequent analysis of consistency and coverage within, between, and across groups. Referencing the approach of Li et al. (2024) and considering the characteristics of the variable values in this study, the 75th percentile, 50th percentile, and 25th percentile are set as calibration anchor points, representing full membership, the crossover point, and non-membership, respectively³⁹. The specific calibration results are shown in Table. 2.

Necessity analysis of individual conditions

Panel data QCA analysis can yield summary consistency, summary coverage, inter-group consistency adjustment distances, and intra-group consistency adjustment distances, with the calculation results shown in Table. 3. Generally, when the consistency level is greater than 0.9, the conditioning variable can be considered a necessary condition for the outcome variable⁵⁵. As seen in Table. 3, none of the conditioning variables are necessary conditions for the outcome. In the dynamic QCA analysis, the consistency adjustment distance threshold is set at 0.2 in this paper. When it is less than 0.2, it indicates high accuracy in summary consistency and coverage. Otherwise, the influence of time and sample clustering needs to be further explored⁵⁶.

First, the analysis of intra-group consistency adjustment distances shows that all the distances are greater than 0.2, indicating a significant regional effect between the six conditions and the outcomes. This is due to the fact that the samples selected in this study are from the seven major urban clusters in the YRB. Due to differences in geographical locations, these regions have distinct resource endowments, industrial structures, and development models. Furthermore, the historical evolution, administrative divisions, policy orientations, and ecological diversity within each urban cluster contribute to significant variations in the performance of these cities across the conditioning variables, leading to higher intra-group consistency distances. In other words, the performance of these cities under certain conditions does not exhibit a universal pattern, but rather is influenced by the complex factors within the YRB, resulting in a highly heterogeneous distribution of conditioning variables⁵⁷.

Second, the analysis of inter-group consistency adjustment distances reveals that the adjustment distances for the two conditioning variables, Innovation Productivity and Environmentally Friendly Productivity, are both less than 0.2, and their summary consistency is below 0.9. This suggests that these two factors are

Causal combination situation	Year											
	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	
X_2 and Y	Consistency	0.818	0.67	0.705	0.705	0.569	0.263	0.306	0.352	0.363	0.457	0.450
	Coverage	0.564	0.502	0.646	0.466	0.525	0.39	0.517	0.616	0.438	0.524	0.578
$\sim X_2$ and $\sim Y$	Consistency	0.296	0.27	0.451	0.347	0.483	0.708	0.718	0.69	0.578	0.546	0.621
	Coverage	0.592	0.428	0.518	0.594	0.528	0.575	0.511	0.43	0.501	0.479	0.543
X_3 and Y	Consistency	0.673	0.46	0.406	0.403	0.538	0.467	0.529	0.662	0.684	0.531	0.589
	Coverage	0.585	0.554	0.628	0.551	0.596	0.447	0.538	0.622	0.536	0.53	0.601
X_5 and Y	Consistency	0.673	0.619	0.605	0.676	0.542	0.51	0.481	0.502	0.304	0.144	0.313
	Coverage	0.512	0.516	0.571	0.455	0.453	0.373	0.496	0.654	0.519	0.575	0.657
$\sim X_5$ and $\sim Y$	Consistency	0.282	0.363	0.353	0.346	0.343	0.392	0.516	0.624	0.744	0.883	0.765
	Coverage	0.436	0.465	0.386	0.57	0.427	0.53	0.501	0.47	0.541	0.485	0.592
X_6 and Y	Consistency	0.876	0.745	0.706	0.689	0.586	0.544	0.471	0.381	0.313	0.357	0.299
	Coverage	0.61	0.564	0.618	0.501	0.514	0.478	0.583	0.664	0.524	0.664	0.677
$\sim X_6$ and $\sim Y$	Consistency	0.375	0.367	0.379	0.446	0.445	0.579	0.667	0.727	0.743	0.802	0.786
	Coverage	0.73	0.568	0.475	0.64	0.517	0.641	0.56	0.454	0.544	0.532	0.542

Table 4. Results of between-group analyses with large distances of adjustment consistency.

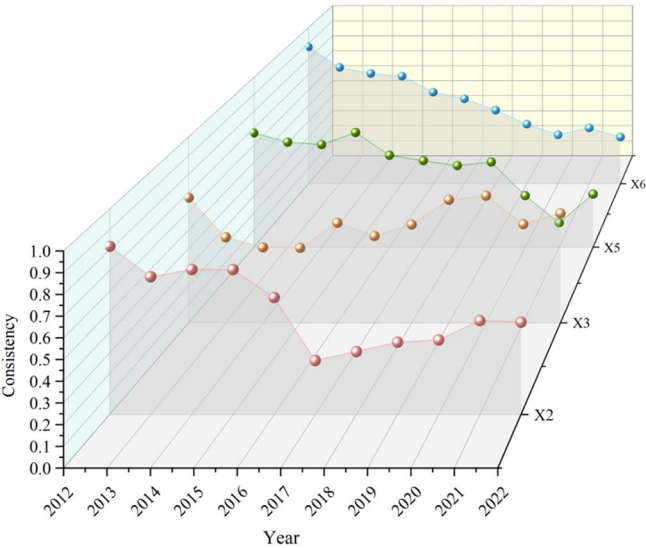


Fig. 5. Variation in intergroup consistency of NQP elements.

not necessary conditions for the outcome variable. However, for the other four conditioning variables—Technological Productivity, Resource-Saving Productivity, Digital Infrastructure Productivity, and Digital Industry Productivity—there are instances where the inter-group consistency is greater than 0.2, warranting further investigation. Given that all preceding conditions are “beneficial factors” for HQED and that “single non-high conditions leading to high outcomes” may be influenced by other confounding factors, the primary focus should be on the relationship between “high-high” and “non-high-non-high” combinations.

The inter-group consistency and coverage of the corresponding variables are shown in Table 4. To observe the necessity of each element of NQP for regional HQED, the variation in consistency over time for each element is plotted in Fig. 5. First, in conditions 1 to 7, the summary consistency across all time dimensions is less than 0.9, indicating that none of these elements pass the necessity test. Second, the consistency level of high resource-saving productivity shows a fluctuating upward trend, suggesting that this factor is gradually becoming a decisive element for HQED. Third, the consistency levels of high technological productivity, high digital infrastructure productivity, and high digital industry productivity all exhibit a clear downward trend, indicating that the necessity of these three factors, when considered individually, is gradually decreasing. Therefore, it is necessary to further focus on the combined effect of these factors with others on HQED.

Sufficiency analysis of configurations

Following the approach of Sheng et al.⁵⁸, the consistency threshold is set at 0.8, the PRI threshold at 0.75, and the case frequency threshold is set at 3. The configurational analysis results are shown in Table 5. It can be seen that the summary consistency for the overall solution leading to HQED is 0.832, which is greater than

Conditional variable	Y				~ Y		
	Configuration path 1	Configuration path 2	Configuration path 3	Configuration path 4	Configuration path 5	Configuration path 6	Configuration path 7
X ₁	●	●	●	●	⊗	⊗	⊗
X ₂	⊗	⊗			⊗		●
X ₃		●	⊗	●	⊗	⊗	●
X ₄		⊗		▲	●	⊗	●
X ₅	⊗		⊗	●	●	●	⊗
X ₆			⊗	⊗		●	⊗
Consistency	0.864	0.855	0.819	0.816	0.868	0.847	0.896
PRI	0.825	0.791	0.729	0.699	0.788	0.766	0.802
Coverage	0.309	0.152	0.182	0.116	0.104	0.155	0.063
Unique coverage	0.100	0.021	0.036	0.031	0.052	0.108	0.038
Intergroup consistency adjustment distance	0.098	0.091	0.120	0.142	0.098	0.080	0.094
Intra-group consistency adjustment distance	0.332	0.368	0.297	0.290	0.269	0.311	0.283
Overall consistency	0.832				0.851		
Overall PRI	0.78				0.782		
Overall coverage	0.413				0.252		

Table. 5. Configuration analysis results. ● and ⊗ represent the presence and absence of core conditions, respectively; ▲ represents the presence of marginal conditions; and blank indicates that both presence and absence are acceptable, i.e., fuzzy conditions.

0.75, indicating that the research results have good explanatory power. The four configurations are sufficient conditions for producing HQED.

(1) Configurational results for HQED

Single-Core Innovation–Driven Model (M_1). This model comprises Configurations 1 and 3, in which innovation productivity consistently appears as a core condition, forming a “second-order equivalent configuration,” while digital infrastructure productivity is absent as a core condition in both. Representative cases include Taiyuan, Zhengzhou, and Yinchuan. These cities prioritize enhancing innovation capacity as their primary development goal, driving economic transformation through technological breakthroughs, product R&D, and the optimization of innovation mechanisms, thereby compensating for relatively underdeveloped digital infrastructure and laying a solid foundation for HQED. The existence of M_1 aligns with the “knowledge spillover” theory, revealing that in the wave of digitalization, a region’s deep technological foundation and strong absorptive capacity can partially offset shortcomings in digital infrastructure⁵⁹—particularly in cities with strengths in industrial technology and engineering sciences. However, without digital empowerment, traditional innovation models may encounter bottlenecks in efficiency and market responsiveness, contrasting with perspectives that emphasize the integration of digital and real economies⁶⁰. The findings of this study suggest that, at least at the current stage, this is a viable path for cities with robust traditional innovation systems.

Region-Wide Innovation Resource Synergy–Driven Model (M_2). This model corresponds to Configuration 2, where innovation productivity and resource-efficient productivity are present as core conditions, while technological productivity and environmentally friendly productivity are absent as core conditions. In this mode, urban development relies not only on technological innovation to drive industrial transformation but also places strong emphasis on resource sustainability. In contrast, this mode does not particularly stress the extensive application of technology or environmental protection, which may be because existing innovation outcomes already meet environmental standards, or because resource efficiency inherently delivers certain environmental benefits. Representative cases include Baotou, Ordos, and Yulin—most of which are resource-based cities in the YRB. This pathway provides strong empirical evidence for the “de-curse” of resource-based cities in the YRB. While traditional views hold that such cities are prone to path dependence and struggle to achieve sustainable development⁶¹, the discovery of the M_2 model is consistent with recent studies showing that “green innovation” can facilitate the transformation of resource-based cities⁶², decoupling economic growth from resource consumption by directly applying technological innovation to enhance resource utilization efficiency.

Comprehensive Digital Transformation–Driven Model (M_3). This model corresponds to Configuration 4 and is the most comprehensive among the four configurations, as well as the only one in which digital infrastructure productivity appears as a core condition. Specifically, the core of digital infrastructure productivity lies in enhancing the YRB’s level of intelligence and digital efficiency, providing fundamental technological support for the generation of NQP. In this configuration, digital infrastructure serves as an “enabler” within the overall driving mechanism, and its combination with innovation productivity and environmentally friendly productivity amplifies the catalytic effect of each core element on HQED, forming a “technology–green–digital” integrated driving logic. Representative cases include Jinan and Xi’an. The findings are highly consistent with research conclusions that digital transformation and green innovation can be mutually enabling and jointly drive high-quality development⁶³. Digital platforms reduce the coordination costs and information asymmetry involved

in green technology R&D, while the demand for green development opens up new market opportunities for technological innovation and digital applications, creating a positive feedback loop for upward growth.

(2) Configurational results for non-HQED

Technology–Green Intensification Deficiency under Digital-Driven Development (S_1). This category includes Configurations 5 and 6. Both pathways indicate that in the process of non-HQED, certain YRB cities share the common characteristic of relying on digital infrastructure and the digital industry as driving forces, while lacking innovation productivity, resource-efficient productivity, and environmentally friendly productivity. Representative cases include Baiyin, Yuncheng, and Dongying. S_1 reveals a potential pitfall of “digital-driven” development—namely, the one-sided pursuit of digital infrastructure and digital industry expansion while neglecting the deep integration with innovation in the real economy and green transformation. This finding resonates with the discussion on the “productivity paradox”⁶⁴. If digital technologies fail to effectively empower innovation processes in core industries, they may instead lead to resource misallocation and a scenario of “growth without quality.”

Innovation–Digital Economy Deficiency under Sustainable Development (S_2). Configuration 7 shows the absence of innovation productivity, digital infrastructure productivity, and digital industry productivity as core conditions, despite some achievements in resource efficiency, environmental protection, and a certain level of technological capability. Representative cities include Dingxi, Tianshui, and Yangquan. The S_2 pathway illustrates another type of development dilemma: although these cities have made certain progress in environmental protection and resource conservation, the absence of innovation and digital economy empowerment renders such sustainability low-level and fragile, making it difficult to achieve truly high-quality development.

Inter-configuration analysis

Examining the trends in inter-configuration consistency over time addresses the limitation of static configuration analysis in traditional QCA studies, overcoming the “temporal blind spot” and revealing the dynamic evolution of multi-factor combination patterns driving HQED in the YRB. As shown in Table 5, the inter-configuration consistency-adjusted distances for all four configurations are below 0.2, indicating no significant time effects. This suggests that the findings of this study have strong applicability to the advancement of HQED under normal conditions. The changes in inter-configuration consistency are illustrated in Fig. 6. Between 2012 and 2022, the consistency trends for all four configurations remained between 0.7 and 1, with relatively small overall fluctuations. Certain time points exhibit a discernible pattern in the fluctuation of inter-configuration consistency, indicating that HQED promotion modes have a strong path dependence. However, a closer examination of the trends in inter-configuration consistency for each condition configuration during 2012–2022 reveals that Configurations 1 and 3 experienced a decline in consistency between 2014 and 2015, with the consistency level of Configuration 3 falling below 0.75. This indicates that during this period, the explanatory power of these two configurations for HQED in the YRB diminished, and the responsiveness of a single innovation element in promoting HQED was weakened. All four configurations saw a notable decline in 2016–2017, with a certain degree of concentration rather than random distribution, suggesting that the decline was not a benign deviation. This period coincided with supply-side structural reforms that led to significant “capacity reduction” in resource-based industries, while stringent environmental policies further constrained the production space for high-pollution, high-energy-consuming industries. As a result, the contribution of traditional industries declined, and innovation-driven and digital economy development had yet to provide effective support, weakening the synergetic effects of various productivity factors. This is consistent with existing research, which suggests that major macro policy

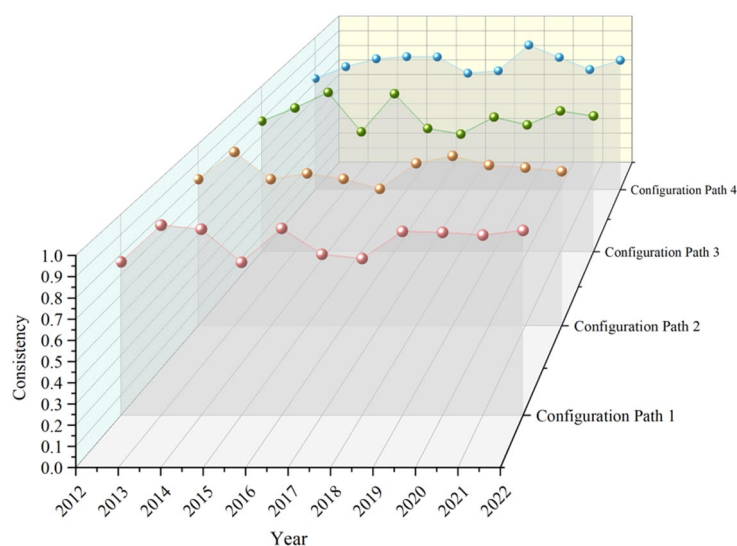


Fig. 6. Changes in intergroup consistency.

shocks can temporarily disrupt or reshape the original industrial structure and development pathways of a region⁶⁵. During 2018–2022, China experienced multiple rounds of tariff hikes and policy confrontations in the Sino–US trade war, which caused severe volatility in financial markets. The subsequent COVID-19 pandemic further disrupted economic activities in China. These issues exhibited time-lag effects, and their adverse impacts have only become evident in recent years.

Intra-configuration analysis

The intra-group consistency for each configuration is shown in Figs. 7 and 8. The performance of the four configurations across cities in the YRB can effectively explain the process of HQED, with the average consistency for each configuration path being above 0.8. Among these, Configuration 4 stands out the most, with an average consistency of 0.831. This indicates that Configuration 4, as a comprehensive driving path, is widely applicable to most cities in the YRB and aligns closely with the region's industrial transformation and the demands for HQED.

The intra-configuration consistency-adjusted distances are all greater than 0.2 (see Table. 4), indicating that the four configurations exhibit preference differences across regions and warrant further examination. To explore the differences in coverage among different urban agglomerations, this study analyzes the spatial distribution of explanatory cases. As shown in Table. 6 and Fig. 9, the four configuration pathways display significant regional variation in the cases they can explain.

Configurations 1 and 3 both correspond to the single-core innovation-driven pathway, with explanatory cases located in the Lanxi Urban Agglomeration and Jinzhong Urban Agglomeration. This indicates that these urban agglomerations benefit from the high concentration of innovation resources and advanced industries, enabling strong innovation capacity and high efficiency in the transformation of scientific and technological achievements. This finding confirms the geographic clustering characteristics of innovation activities⁶⁶. However, this model can exacerbate unbalanced development in the YRB, as the economic and technological capacity gap between core cities and surrounding areas may further widen. Therefore, greater attention should be paid to the radiating effects of core cities to drive HQED in neighboring regions. Configurations 2 and 4 represent two types of comprehensive development models. Compared with the single-core innovation-driven pathway, these configurations are more suitable for urban agglomerations with a solid industrial base, high ecological sensitivity, and relatively advanced resource management and digitalization capabilities. Their explanatory cases are located in the Shandong Peninsula Urban Agglomeration and the Central Plains Urban Agglomeration. The development advantage of these regions lies in their relatively balanced economic structures, which have fostered complementary industrial chains and innovation cooperation networks. As noted by Frenken et al.⁶⁷, the coexistence of technologically related and diversified industries within a region is more conducive to cross-domain knowledge spillovers and the emergence of new industries, thereby enhancing regional economic resilience and long-term growth potential. Consequently, the comprehensive development model can be viewed as one in which an effective innovation network forms within the urban agglomeration, enabling the coordinated synergy of technological, green, and digital factors to jointly promote high-quality regional development.

Discussion

The core contribution of this study is to demonstrate that HQED in the YRB is not a linear process driven by a single factor, but rather a complex adaptive process profoundly shaped by historical path dependence and external shocks. The identified “W-shaped” evolutionary trend—unlike the trajectories observed in the Beijing–Tianjin–Hebei, Yangtze River Delta, and Pearl River Delta urban agglomerations^{51,52}—is not a simple cyclical fluctuation. Instead, it reflects the vulnerability of endogenous growth momentum and the pains of structural

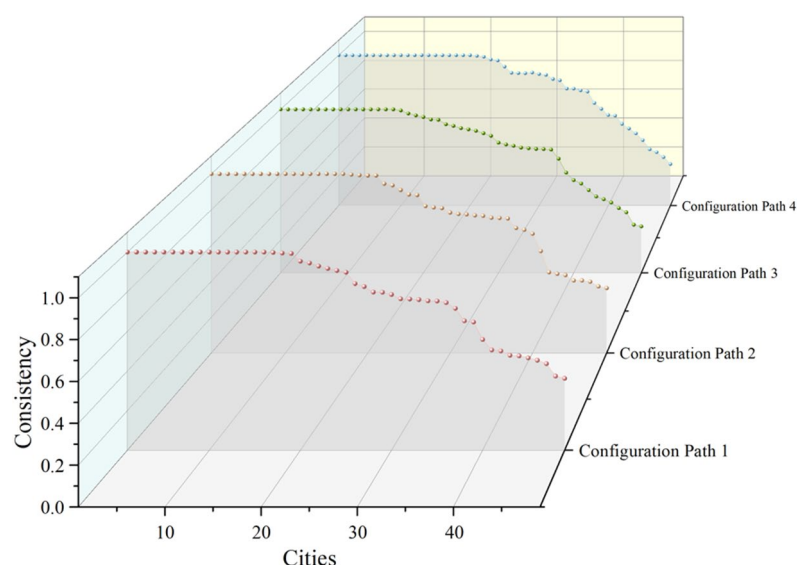


Fig. 7. Intra-group consistency of each configuration.

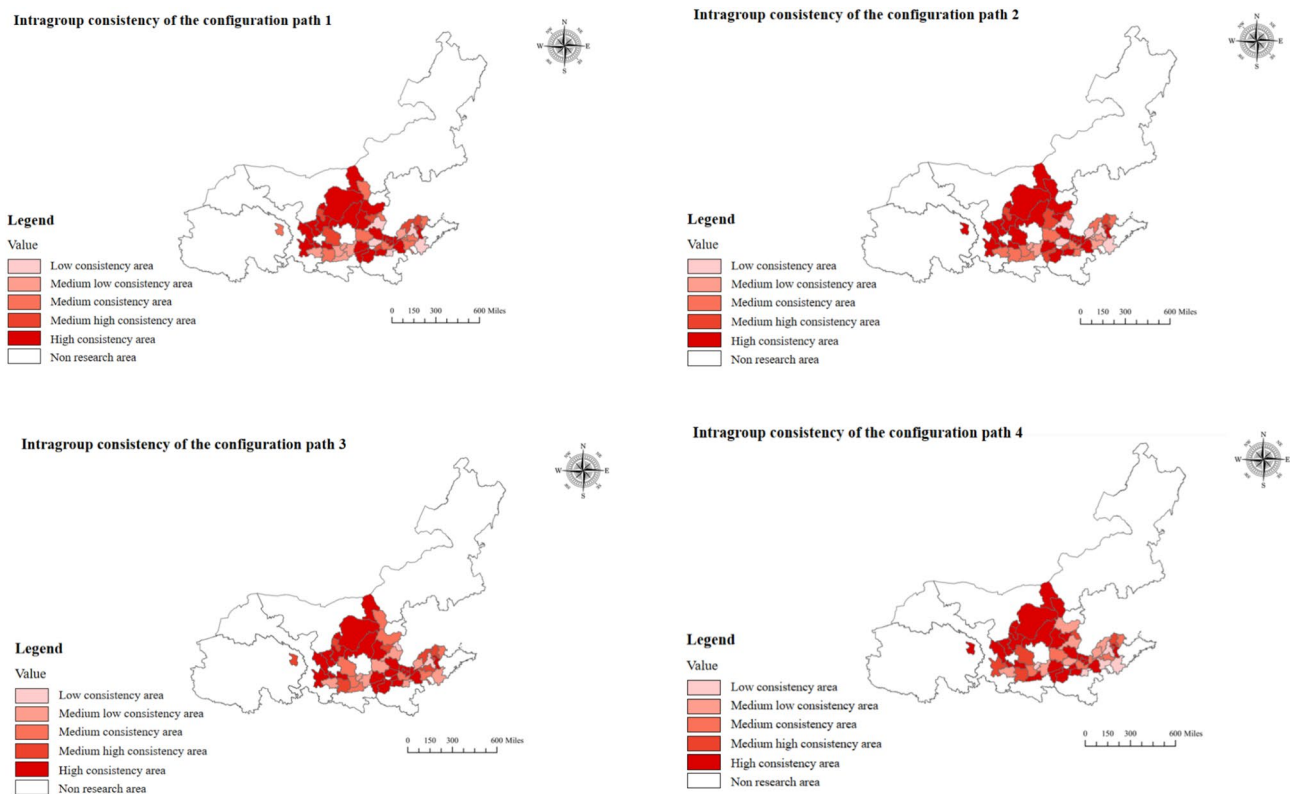


Fig. 8. Spatial distribution of intra-group consistency for each configuration.

Urban Cluster	Configuration path 1	Configuration path 2	Configuration path 3	Configuration path 4
Central plains urban cluster	0.221	0.186	0.117	0.168
Lanxi urban cluster	0.460	0.160	0.239	0.036
Guanzhong plains urban cluster	0.220	0.155	0.211	0.135
Hubao-Yuzhou-eqiu urban cluster	0.185	0.096	0.173	0.049
Ningxia yellow river urban cluster	0.192	0.112	0.105	0.084
Shandong peninsula urban cluster	0.398	0.226	0.371	0.189
Jinzhong urban cluster	0.424	0.155	0.335	0.060

Table. 6. Coverage of the seven YRB city cluster groupings.

transformation in the YRB, a typical inland river basin, when confronted with successive external shocks such as supply-side structural reforms, Sino-US trade frictions, and the COVID-19 pandemic. This finding deepens Zhang et al.’s (2022) perspective on the staged nature of regional transformation, emphasizing the nonlinearity and high uncertainty of the process, and challenging traditional views that regard economic transition as a smooth progression⁶⁸. HQED in the YRB also exhibits a distinct “higher in the east, lower in the west” spatial gradient. This spatial heterogeneity stems from historically accumulated differences in industrial foundations, disparities in factor endowments, and variations in geographical location. The middle and upper reaches of the YRB are primarily agricultural and pastoral areas⁶⁹, facing severe challenges such as frequent extreme droughts and floods, as well as intensifying conflicts between water supply and demand⁷⁰, all of which hinder economic progress. Moreover, as a critical “ecological barrier zone,” the upper reaches of the YRB have suffered substantial ecological losses from conservation and restoration efforts, but the absence of adequate ecological compensation has further constrained their economic development⁷¹.

This paper constructs the “Technology-Green-Digital” three-dimensional NQP theoretical framework, which not only expands the research paradigm of Marxist theory of productive forces but also provides a new interpretative perspective for understanding HQED in the YRB in the digital age. Unlike traditional productivity theory, which emphasizes the input of material capital and labor, NQP places greater emphasis on the importance of knowledge capital, digital technologies, and ecological factors³⁴. This theoretical extension aligns with Romer’s (1990) endogenous growth theory, which discusses the importance of the knowledge economy, and further incorporates modern elements of sustainable development and digital transformation⁴⁵. It thus provides a new interpretation of the internal and external driving factors of high-quality development in the YRB.

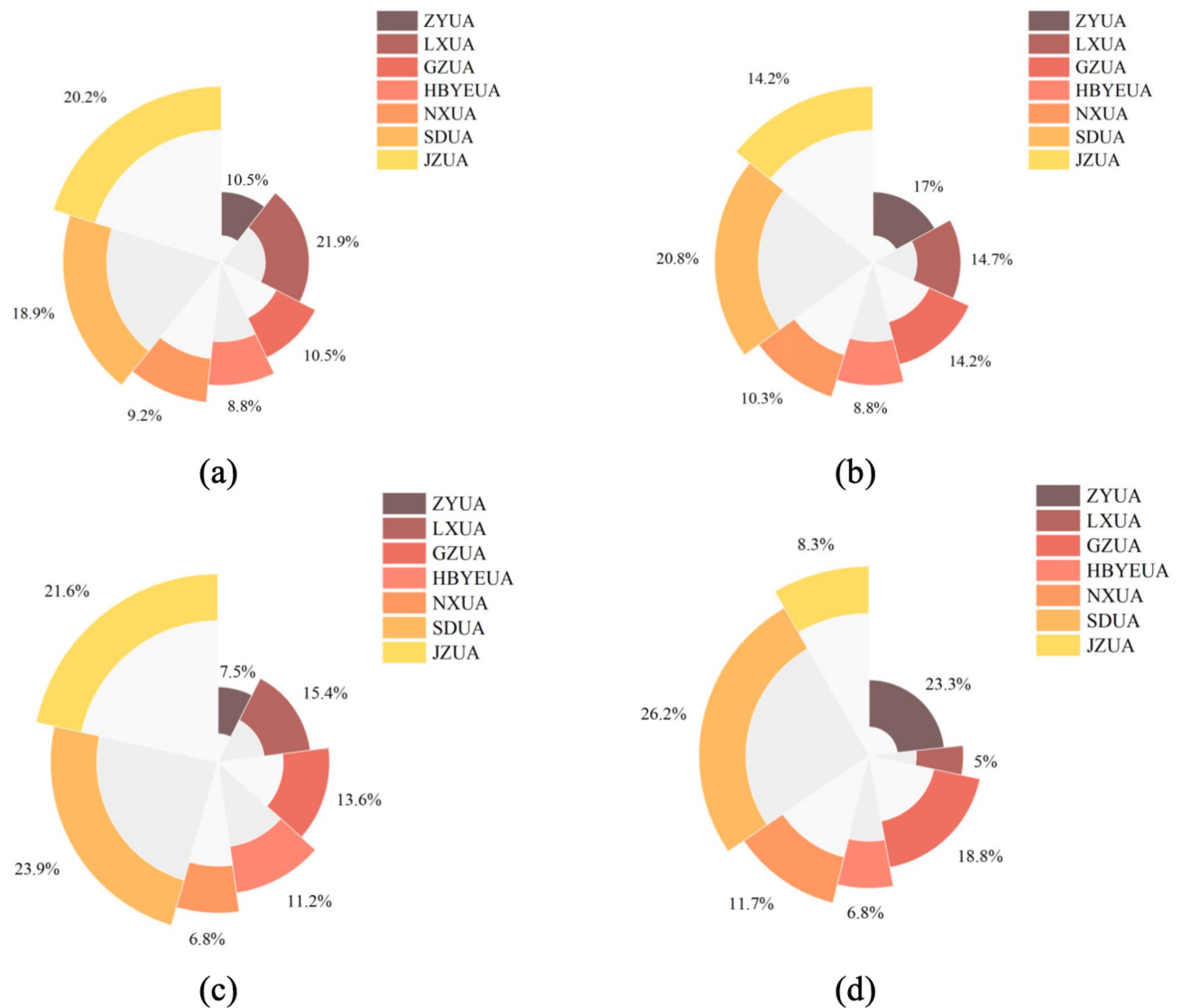


Fig. 9. YRB's seven city cluster grouping coverage.

At a deeper level, the three configurational pathways of HQED identified in this study provide key economic-geographical insights into how different regions explore development models under the NQP framework. First is the “single-core innovation-driven” pathway, concentrated in the Lanxi and Jinzhong urban agglomerations. This reflects a typical administrative-regional economic logic⁷². Provincial capital cities in these areas concentrate the majority of scientific research and educational resources, creating “innovation highlands.” However, under conditions of underdeveloped market mechanisms and a weak digital economy base, innovation elements struggle to diffuse outward, resulting in “innovation islands” disconnected from local industries²⁷. Second is the “region-wide innovation resource synergy-driven” pathway, commonly found in resource-based cities. This can be viewed as a crisis-driven green transformation model⁷³. Confronted with the dual pressures of capacity reduction and environmental regulation, survival crises compel these cities to allocate their limited innovation resources precisely toward improving resource efficiency and achieving environmental friendliness⁷⁴. This finding suggests a potential solution to the “resource curse” theory: centering on green technological innovation may guide resource-based cities onto an endogenous path of sustainable development, echoing the arguments of Si et al.⁶². Third is the “comprehensive digital transformation-driven” pathway, more prevalent in the Central Plains and Shandong Peninsula urban agglomerations. This is due to their more diversified industrial structures, broader market hinterlands, and stronger digital foundations^{75,76}. It confirms the effectiveness of integrating “technology–green–digital” elements. At the same time, such integration is itself a product of path dependence, requiring a relatively balanced and advanced economic system as fertile ground for its development.

A core insight emerging from these three equifinal pathways is the dual role of innovation productivity: it is both an indispensable universal driver across all models of high-quality development and a force whose effectiveness is highly contingent on contextual synergy^{77,78}. The universality of innovation productivity lies in its role as the fundamental source of breaking old production functions and creating new value in the knowledge

economy era, consistent with the central arguments of endogenous growth theory⁴². However, the findings of this study decisively refute any linear form of “innovation determinism.” The very existence of three distinct pathways powerfully demonstrates that the vast potential of innovation is intrinsic, and whether—and how—it can be transformed into real productive forces depends entirely on the specific regional configuration into which it is embedded. In resource-based cities, innovation is channeled by survival pressures toward green development; in administratively dominated regions, innovation remains constrained in “isolated islands” due to the lack of digital empowerment; and only within comprehensive regional systems can innovation fully resonate with digital and green elements, unleashing its greatest multiplier effect. Accordingly, this study positions innovation as a “catalyst”—its transformative power must be unlocked and shaped by the particular regional configuration in which it operates.

Finally, unlike existing QCA studies that primarily focus on analyzing static relationships between antecedents and outcomes^{79,80}, this study employs dynamic QCA to reveal that spatial differentiation is significantly greater than temporal variation. This conclusion provides robust empirical support for the theories of path dependence and “lock-in” effects in evolutionary economic geography. The historical foundations of the YRB’s urban agglomerations exert a strong lock-in effect on their current development models. This implies that breaking the entrenched “higher in the east, lower in the west” pattern and achieving basin-wide coordinated development cannot rely solely on spontaneous market forces; rather, it requires higher-level institutional designs and interventions aimed at helping specific regions “unlock” their existing paths. In sum, this study offers profound insights into understanding the complexities of regional transformation in China.

Conclusions and recommendations

Conclusion

Through a dynamic configurational analysis of HQED in the YRB, this study reveals that NQP is not a linear aggregation of single factors but rather a complex system driven by the synergy of multiple elements. The findings confirm that HQED in the YRB does not follow a steady growth trajectory; instead, its “W-shaped” evolutionary pattern and pronounced “higher in the east, lower in the west” spatial gradient together reflect the nonlinear development characteristics typical of China’s inland river basins under the dual influences of macro policy shocks and structural transformation.

More importantly, the empirical results reject the commonly held “single-factor determinism” in regional development. The study demonstrates that no individual NQP element constitutes a necessary condition for HQED. This conclusion advances the research paradigm from searching for “the most important driver” to identifying “effective combinations of factors,” underscoring the necessity of adopting a configurational perspective in the study of complex regional issues. On this basis, the study identifies three equifinal core pathways: the single-core innovation-driven model, the region-wide innovation resource synergy-driven model, and the comprehensive digital transformation-driven model. This finding provides strong empirical evidence for the principle of equifinality in regional development. It challenges “one-size-fits-all” policy approaches, offering diverse and feasible blueprints for cities with varying resource endowments. It also highlights that while technological innovation plays a universal role, its effectiveness can only be maximized when operating in synergy with other factors.

Finally, the study finds no significant temporal effects in the configurational pathways, but clear spatial heterogeneity is evident. The Central Plains and Shandong Peninsula urban agglomerations tend to rely on comprehensive development models, while the Lanxi and Jinzhong agglomerations depend more heavily on single-core innovation-driven models. This phenomenon confirms the strong path dependence of regional development models and demonstrates the critical role of geographical location and historical foundations in shaping development paradigms. These insights provide valuable evidence for economic geography and regional policymaking, emphasizing that “space outweighs time” in understanding regional evolution.

Recommendations

Building on the identified diverse development pathways and pronounced spatial heterogeneity, advancing HQED in the YRB requires shifting policymaking from broad macro-level exhortations to configuration-based targeted interventions. This entails designing differentiated development strategies tailored to the path-dependent characteristics of each urban agglomeration and translating the principle of “equifinality” into place-based action plans. For the Lanxi and Jinzhong urban agglomerations, which follow the “single-core innovation-driven” pathway, policy priorities should focus on deepening the integration between innovation chains and local industrial chains. Specific measures may include establishing dedicated funds and tax incentives to encourage traditional local enterprises to adopt scientific and technological achievements from core cities, thereby preventing innovation resources from becoming isolated “islands” disconnected from local economic development. For resource-based cities such as Baotou and Ordos that follow the “region-wide innovation resource synergy-driven” pathway, policies should concentrate on incentivizing green technological innovation with resource conservation at the core. This could involve creating special green technology funds for the transformation of resource-based cities, guiding them to leverage technological innovation to overcome the “resource curse.” For the Central Plains and Shandong Peninsula agglomerations, which exhibit the “comprehensive digital transformation-driven” pathway, policy efforts should aim at building demonstration zones for the integration of technology, green development, and digitalization across the basin. These zones could be supported in pioneering projects such as smart energy networks while assuming the role of core engines that disseminate technologies, standards, and management experience to other regions within the basin.

To address the overall “W-shaped” fluctuations and the entrenched “higher in the east, lower in the west” spatial imbalance in HQED, the successful implementation of these differentiated strategies must be underpinned by a new basin-wide coordination mechanism that transcends administrative boundaries and

targets configuration complementarities. The construction of this mechanism should proceed on two levels. Functionally, traditional ecological compensation should be upgraded into benefit-sharing for development. For example, a Yellow River Basin Coordinated Development Fund could be established, financed partly by tax revenues generated from transferring downstream industries upstream. The use of this fund should expand beyond ecological protection to include financing upstream infrastructure development and workforce training necessary to accommodate industrial relocation. Operationally, cross-agglomeration project collaborations based on “advantageous configurations” should be actively promoted and financially supported. For instance, cities with comprehensive digital strengths (e.g., Xi’an) could jointly develop smart mining and digital energy projects with cities strong in single-core innovation (e.g., Jinzhong), or the Shandong Peninsula could leverage its capital and market advantages to incubate technological achievements in the Lanxi agglomeration. Such project-oriented collaboration can transform abstract notions of regional coordination into concrete, productivity-optimizing arrangements that fully capitalize on each region’s configurational strengths, thereby enhancing the resilience and synergetic capacity of development across the entire YRB.

Research limitations and prospects

This study still has some limitations that should be addressed in future research:

Empirical Analysis Based on Macroscopic Data: This study’s empirical analysis mainly relies on macro-level statistical data at the city level, which makes it difficult to deeply reveal the interactive mechanisms between micro-level entities (such as businesses and research institutions) and how NQP elements can be effectively coordinated at the micro level. Future research could combine survey data and case studies to explore the relationship between micro-level mechanisms and macro-level patterns, providing a more solid micro-foundation for NQP theory.

Limited Sample Scope: The sample in this study is limited to 49 prefecture-level cities in the YRB. While this provides some regional representativeness, the external validity of the research findings requires further verification. Future studies could expand the sample range to include cross-basin comparative research or conduct international comparisons with cities from developed countries to test the universal patterns and specific pathways through which NQP influences HQED.

Lack of In-depth Exploration of Mediating Mechanisms: This study primarily focuses on the relationship between NQP and HQED, but the exploration of the mediating mechanisms between them is not sufficiently in-depth. Future research could introduce mediating variables such as industrial upgrading and institutional innovation, and construct a more complete theoretical model to deepen the understanding of the intrinsic mechanisms through which NQP promotes HQED.

Data availability

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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Author contributions

Y.S. validated results, provided resources, acquired funding, reviewed/edited the manuscript. X.L. collected and processed data, implemented software, created visualizations, and wrote the original draft, and served as corresponding author. J.Z. conceived and designed the study, developed the methodology, performed formal analysis, supervised the research, and reviewed/edited the manuscript. All authors contributed to the final version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Ethical approval

All authors have read, understood, and have complied as applicable with the statement on “Ethical responsibilities of Authors”.

Additional information

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